

AI-Powered Credit Risk Intelligence Platform

An end-to-end credit-risk solution on the Home Credit Default Risk dataset — calibrated ML scoring, SHAP explainability, model-derived business rules, and an LLM natural-language → SQL interface, delivered as one interactive application.

307,511

APPLICANTS

0.763

ROC-AUC

5

INTEGRATED MODULES

1

DOCKER COMMAND

Credit decisions have to be accurate *and* explainable

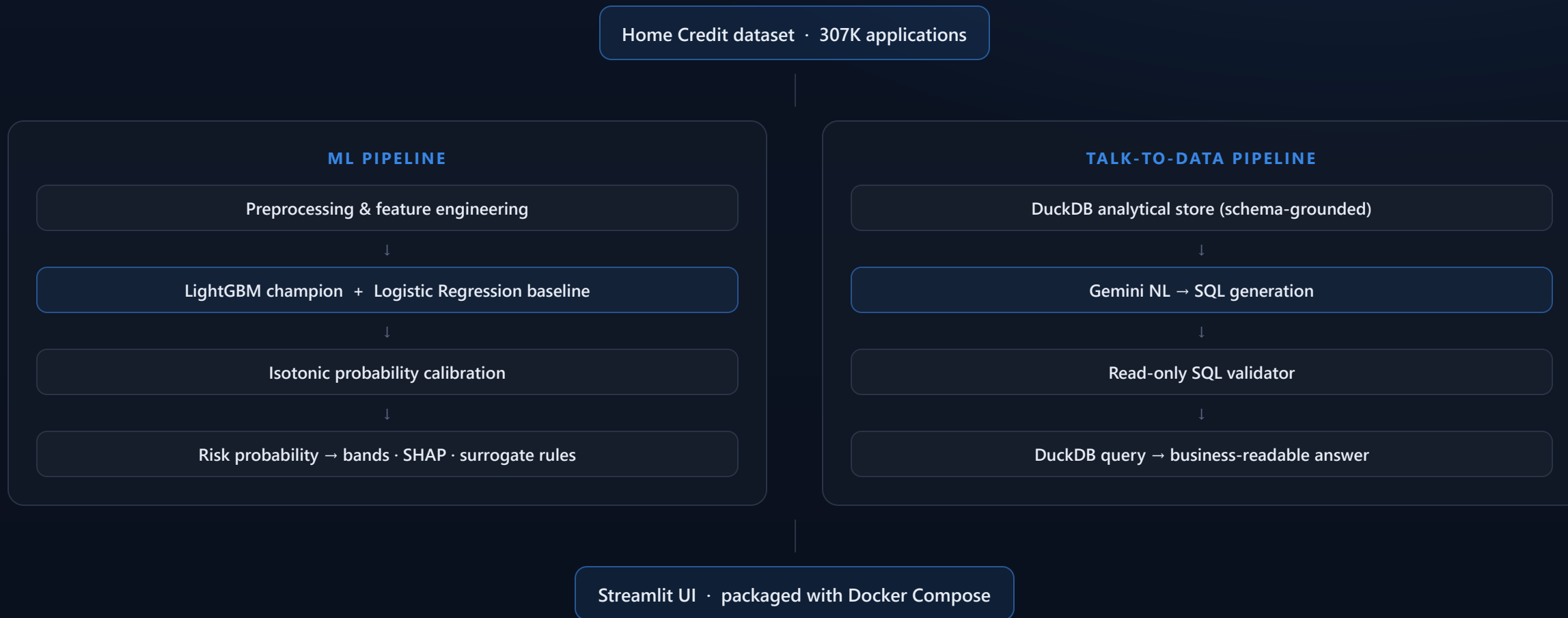
Lenders need default-risk assessment that survives credit-committee review, internal audit, and regulatory scrutiny. A single black-box score does not — the decision has to be traceable and usable by non-technical staff.

- Risk teams need a **defensible number** and the reasons behind it
- Analysts need **readable policy**, not model internals
- Business users need to **interrogate the data** without SQL

WHAT THE PLATFORM DELIVERS

- Calibrated default probability + Low / Medium / High band
- Per-applicant SHAP reason codes
- Human-readable risk rules for analyst review
- Plain-English querying of the applicant data
- One Streamlit app, reproducible via Docker Compose

System architecture

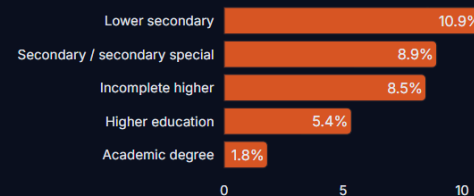


307K applications, ~8% default — a hard class imbalance

- **External credit scores** are the strongest default signal — 19.9% → 2.6% across EXT_SOURCE bands
- Default rate **falls steadily with age** (18–25 highest)
- **Education** separates risk: 10.9% lower-secondary vs 1.8% academic degree
- Credit-to-income ratio is **weak alone**, useful combined
- ~19% of rows carry a "not currently employed" placeholder — handled as an explicit flag
- **Heavy missingness** — up to ~70% in housing columns

Business insights

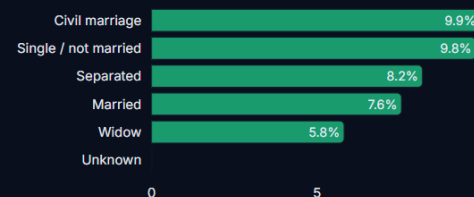
Default rate by education level



Default rate by age



Default rate by family status



Default rate by EXT_SOURCE_2 decile

Low → high external credit score.



LightGBM champion, calibrated to a real probability

MODEL	ROC-AUC	PR-AUC
Logistic Regression (baseline)	0.750	0.235
LightGBM + isotonic calibration	0.763	0.246

- At ~8% positives, **accuracy misleads** — ROC-AUC & PR-AUC drive selection
- **Isotonic calibration** anchors scores to the true base rate, so the probability reads as a real rate
- Bands off the **calibrated probability**: Low < 0.20 · Medium 0.20–0.50 · High ≥ 0.50
- Scoring path is **version-proof** — calibrator unwrapped and re-applied explicitly



Every score ships with its reasons

- SHAP TreeExplainer on the LightGBM model
- Per-applicant signed contributions — red raises risk, blue lowers it
- Global mean-|SHAP| importance over a 1,000-applicant sample
- EXT_SOURCE features **dominate** both local and global views
- Framed for **analysts**, not only data scientists

WHY THIS SCORE

Explainability

Which features drove the model's score — SHAP, per applicant and global.

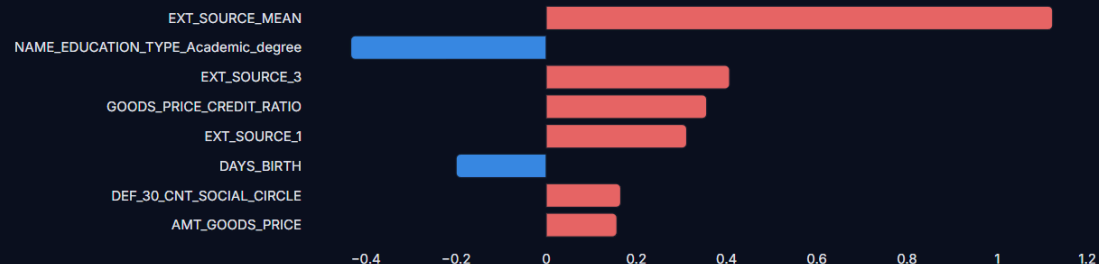
PREDICTED DEFAULT PROBABILITY

42.4%

Top factors for this applicant

Signed SHAP contribution — red raises the predicted risk, blue lowers it.

← lowers risk | raises risk →



A human-readable approximation of the model

- Shallow **surrogate decision tree** (depth ≤ 3) trained to mimic the champion
- Each leaf is an **IF / AND rule** with its observed default rate — 2.4% → 34.1%
- Companion table: observed default rate per **EXT_SOURCE band**, against real TARGET labels
- Rules that **fire for the scored applicant** are listed
- Explicitly **decision support**, not lending policy

Business Rules

Human-readable credit policy derived from the model — for analyst review and audit.

Policy rules

Surrogate decision tree (depth ≤ 3) trained to mimic the champion.

IF EXT_SOURCE_MEAN $\leq$ 0.395 AND EXT_SOURCE_MEAN $\leq$ 0.25 AND EXT_SOURCE_3 $\leq$ 0.226 384 applicants	34.1%
IF EXT_SOURCE_MEAN $\leq$ 0.395 AND EXT_SOURCE_MEAN $\leq$ 0.25 AND EXT_SOURCE_3 $>$ 0.226 832 applicants	23.3%
IF EXT_SOURCE_MEAN $\leq$ 0.395 AND EXT_SOURCE_MEAN $>$ 0.25 AND GOODS_PRICE_CREDIT_RATIO $\leq$ 0.826 793 applicants	20.6%
IF EXT_SOURCE_MEAN $\leq$ 0.395 AND EXT_SOURCE_MEAN $>$ 0.25 AND GOODS_PRICE_CREDIT_RATIO $>$ 0.826 2,383 applicants	14.1%
IF EXT_SOURCE_MEAN $>$ 0.395 AND EXT_SOURCE_MEAN $\leq$ 0.523 AND GOODS_PRICE_CREDIT_RATIO $\leq$ 0.863 2,105 applicants	10.2%
IF EXT_SOURCE_MEAN $>$ 0.395 AND EXT_SOURCE_MEAN $\leq$ 0.523 AND GOODS_PRICE_CREDIT_RATIO $>$ 0.863 3,509 applicants	7.1%

Ask the applicant data in plain English

- Schema-grounded prompt → Gemini generates SQL
- Read-only validator — DROP / DELETE / UPDATE / INSERT / ALTER rejected, approved tables only, row cap
- Pre-verified templates answer common questions with no LLM call
- Conversation memory resolves follow-ups ("what about women only?")
- Generated SQL + result table always shown for transparency

Sample questions

One click — pre-verified SQL, no LLM call.

What is the overall default rate?

What is the average income of applicants who defaulted vs did not?

Show default rate by education level

How many applicants own both a car and a house?

What are the top 5 organization types by number of applicants?

What is the overall default rate?

Returned 1 row(s) across columns: default_rate_pct, n_applicants.

SQL used

```
SELECT AVG(TARGET) * 100 AS default_rate_pct, COUNT(*) AS n_applicants FROM applications
```

default_rate_pct	n_applicants
8.0729	307511

Ask about the applicant data...

Limitations & what's next

CURRENT LIMITATIONS

- Application table only — no bureau / previous-loan aggregates
- Calibration uses the evaluation fold
- Hyperparameter search intentionally limited
- Free-form NL → SQL needs an external API key
- Rules approximate the model — not formal policy
- No fairness audit or production banking validation

NEXT STEPS

- Aggregate bureau & previous-application history into features
- Automated hyperparameter optimization
- Dedicated train / calibration / test splits
- Model-drift monitoring
- Fairness & bias evaluation
- API inference service with authentication

SUMMARY

Built with a lean, reproducible stack

LAYER	TECHNOLOGY
Data processing	Python · Pandas · NumPy
Analytical database	DuckDB
Machine learning	scikit-learn · LightGBM
Explainability	SHAP
LLM · SQL safety	Google Gemini · sqlparse
UI · visualization	Streamlit · Plotly
Deployment	Docker · Docker Compose

IN ONE LINE

Data → EDA → feature engineering → calibrated ML → explainability → business rules → NL-to-SQL → UI → Docker — the full AI-engineering lifecycle in one app.

Analytical prototype for credit-risk decision support. Not a substitute for formal lending policy, fairness auditing, or regulatory approval.